

Our R&D and training related activities go hand in hand and the students can benefit a lot from that. ”

—Shahim Baker, Director (Operation), Grey Technolabs

trying to bridge the gap between the industry and academia. The role of the academic institution ceases soon after the course ends.

“We see a brighter scope of moving up the value chain where institutions with strong domain expertise can take up innovation/technology incubation programme in collaboration with the industry. This is more feasible at the training institutes’ end as they have better touch with the academia and act as an interface with the industry,” says Raju.

Moreover, the industry is crippled with an unorganised market structure, lack of investments, geographical bounds, lack of trained manpower and costs required for constant technological upgradation. If colleges tie up with training institutes, it would be beneficial for the students to groom themselves as an embedded engineer taking infrastructural help from the college and training help from the training institute.

Raju suggests, “It would be wise if institutes try to increase their reach by upgrading to virtual classroom based programmes with trained lab coordinators. This has to be done professionally and on a long-term basis with support staff being trained in parallel.”

Training centres need to become equipped with updated tools and software. “The availability of open source resources (such as Linux and Eclipse) is

a great benefit for the academia to leverage the huge knowledge base of the user community and to impart high-end training at a lower cost,” says Pillai.

A parallel market of immense interest is the other Asian countries. “Owing to the affordability, there is an existing flow of students/professionals from south Asia to Chennai. So this potential market can be taken to a higher level

bling them to engage more on practical level. Jakuku feels, “The trainers need to be re-trained. Moreover, they must be better compensated, thereby raising the quality of people employed. Currently, we choose trainers from the experienced workforce involved in our product development centre.”

Jatinder Singh, CEO, TICO, says that updation of the curriculum in accordance with the fast changing demands of the industries is the key to survival. He puts emphasis on code optimisation, hardware design of mixed-signal products, finding best resources and environments for cost-effective product solutions, and time-to-market implementation in product design.

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—Maaz Jakuku, Director, Emrtex

with global audience,” says Raju.

“Due to the lack of sufficient number of qualified trainers, the Indian institutes are at a basic level in the value chain. The different value-addition stages can be identified as entry level (covering undergraduate courses), intermediate level (presently achieved by training institutes with generic topics in a focused domain), advanced level (to be achieved by the training institutes with specialised in-depth topics on par with the industrial activity),” says Pillai.

Organisations like Emrtex prefer to move up the value chain by training people more and more hands-on, ena-

Kalra lists out quality training, proper infrastructure, students’ awareness about latest tools, focus on practical aspects of the technologies and industry partnership as the key elements for upward movement in the value chain.

Emerging trends

The training providers are shifting towards an online mode of training to avail the benefits of cost reduction and better accessibility.

From the perspective of training topics, the key areas are different verification methodologies like platform-based verification, layered test bench, low-power design methodologies, mobile and handheld device technologies, open verification methodology (OVM) for SoCs, and design and verification using System Verilog.

“A trend is emerging to have a common language for both design as well as verification so as to reduce the overall time to develop an IP. System Verilog is emerging as the favourite

ARM preprocessors and open source embedded software tools such as Linux, Android and AVX will be the sought-after topics of training courses in the future. ”

—Parswanath, Managing Director, Eilabz